Single-Run Scans over Albedo, Asymmetry, and Optical Depth in MoCafe

Abstract

MoCafe can record a peeling-off scattered image for an entire grid of dust parameters from a *single* Monte-Carlo run. Photons are propagated with one simulated asymmetry factor g_0 and one simulated optical depth τ_0, while closed-form per-photon weights—requiring no additional random numbers and no re-tracing—fill in the results for a grid of albedos a, asymmetry factors g, and optical depths τ. The scattered image becomes the five-dimensional array $J(x, y, a, g, \tau)$. The albedo/asymmetry reweighting follows Seon (2010); the optical-depth ("polychromatic") reweighting follows Jonsson (2006). This memo collects the underlying estimator, the reweighting formulae, the proof that the forced first-scattering normalization cancels, and the resulting separable form of the image accumulation.

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1. Baseline Monte-Carlo estimator

MoCafe solves dust radiative transfer with a next-event ("peeling-off") estimator (Yusef-Zadeh, Morris & White 1984): at every emission and scattering site a deterministic estimate of the flux that would reach each observer is added to the image, in addition to advancing the random walk. The forward walk uses two standard variance-reduction devices:

- **Forced first scattering** (Cashwell & Everett 1959). Let $\tau^{\rm e}$ be the optical depth from the emission point to the edge of the grid along the initial direction. The packet weight is multiplied by $(1 - {\rm e}^{-\tau^{\rm e}})$—the probability of interacting before escaping—and the first interaction optical depth is drawn from the truncated exponential on $[0, \tau^{\rm e}]$,

$$\tau_1 = -\ln\left[1 - R(1 - e^{-\tau^{\rm e}})\right], \qquad R \sim U(0,1), \tag{1}$$

whose probability density is

$$f_1(\tau_1) = \frac{e^{-\tau_1}}{1 - e^{-\tau^{\rm e}}}, \qquad \tau_1 \in [0, \tau^{\rm e}]. \tag{2}$$

- **Continuous absorption (reduced weight).** Beyond the first interaction, free paths are drawn from the unbiased exponential, $\tau_k = -\ln R$ with density $e^{-\tau_k}$, and the packet is never killed by absorption; instead its survival is carried as a weight. The packet terminates only by escaping the grid.

Write the packet weight after the forced first scattering as $w \equiv (1 - e^{-\tau^{\rm e}})$ (taking the emission weight to be unity). Let P_k be the position of the k-th scattering, $r_k = |P_k - \mathbf{x}_{\rm obs}|$ the distance to the observer, $\tau_{\rm obs}(P_k)$ the optical depth from P_k to the edge along the observer direction, and $\mu_{{\rm obs},k}$ the cosine of the angle between the incoming direction and the observer direction. For a single dust model with albedo a and asymmetry g, the peeling-off contribution of the k-th scattering to the image pixel onto which P_k projects is

$$\Delta J_k \;=\; \frac{w}{r_k^2}\, a^k\, \frac{\Phi(\mu_{{\rm obs},k}\,|\,g)}{4\pi}\, e^{-\tau_{\rm obs}(P_k)}, \tag{3}$$

where the factor a^k accounts for k survival events (the packet has scattered k times by event k), and Φ is the scattering phase function. The direct (unscattered) image is the analogous estimate evaluated once at the emission point.
MoCafe uses the Henyey–Greenstein (1941) phase function,

$$\Phi(\mu\,|\,g) \;=\; \frac{1 - g^2}{\left(1 + g^2 - 2g\mu\right)^{3/2}}, \tag{4}$$

normalized so that $\int \Phi(\mu|g)\,d\Omega/4\pi = 1$ (here $\mu = \cos\theta$ is the cosine of the scattering angle).

2. Albedo and asymmetry scan (Seon 2010)

The goal is to obtain $J(x,y,a,g)$ on a grid $\{a_i\} \times \{g_l\}$ from one run. The key observation is that the random walk's *geometry*—the interaction points and directions—can be sampled once and reweighted, because neither a nor g changes the extinction optical depth that drives the free paths.

2.1 Asymmetry reweighting

Sample the actual scattering cosines μ_j from the single simulated phase function $\Phi(\cdot\,|\,g_0)$. A path realized with g_0 represents a path with output asymmetry g if each scattering is reweighted by the ratio of phase functions (Seon 2010, eq. 2),

$$W_g^{(k)} \;=\; \prod_{j=1}^{k} \frac{\Phi(\mu_j\,|\,g)}{\Phi(\mu_j\,|\,g_0)}. \tag{5}$$

At the peel of event k the scattering *toward the observer* contributes $\Phi(\mu_{{\rm obs},k}|g)$ directly, while the $k-1$ *previous* scatterings are represented by $W_g^{(k-1)}$. The method is accurate while $|g - g_0|$ is not too large; Seon (2010) finds $g_0 \simeq 0.4$–0.5 optimal for covering $g = 0.0$–0.9.

2.2 Albedo reweighting

The albedo controls only survival, not the extinction optical depth, so the a-grid is filled by replacing the single factor a^k in (3) with a per-albedo accumulator

$$A_a^{(k)} = a^k, \tag{6}$$

updated once per scattering (before the peel, so that the k-th peel sees a^k).

2.3 Four-dimensional image

Combining (3), (5), and (6), the k-th scattering deposits, for every grid point (a_i, g_l),

$$J(x, y, a_i, g_l) \mathrel{+}= \frac{w}{r_k^2} \, \mathrm{e}^{-\tau_{\rm obs}(P_k)} \, A_{a_i}^{(k)} \, \frac{\Phi(\mu_{{\rm obs},k} \,|\, g_l)}{4\pi} \, W_{g_l}^{(k-1)}. \tag{7}$$

The expensive geometric quantities—$\tau_{\rm obs}$ (a ray trace), $1/r_k^2$, and the image pixel—do not depend on (a, g) and are computed *once* per event; only the closed-form factors vary across the $n_a \times n_g$ grid. Because no random numbers are consumed by the reweighting, the slice at (a, g_0) is bit-for-bit identical to an ordinary run with that albedo and $g = g_0$. The direct image is independent of a and g and stays two-dimensional.

3. Optical-depth (polychromatic) scan (Jonsson 2006)

The optical-depth scan is the polychromatic algorithm of Jonsson (2006) specialized to a *grey* rescaling of one medium: all opacities are scaled by a constant factor, so every optical depth along every ray scales by the same amount. Let the simulated reference be τ_0 = `par%taumax` and let the target optical depths be $\{\tau_t\}$, with scaling factors

$$s_t = \frac{\tau_t}{\tau_0}. \tag{8}$$

Because the medium is uniformly rescaled, the optical depth to *any* point along a fixed ray satisfies $\tau^{(t)} = s_t \, \tau^{(\rm ref)}$.

3.1 Free-path reweighting

Consider a segment in which the reference run draws free-path optical depth τ_k and interacts at the physical point P_k. Under the scaling s_t the same physical point carries optical depth $s_t \tau_k$, and the probability of interacting there changes. In the integration variable τ_k, the reference interaction density is $\mathrm{e}^{-\tau_k}$, while the target density is $\mathrm{e}^{-s_t \tau_k} \, s_t$ (the factor s_t is the Jacobian $\mathrm{d}(s_t \tau_k)/\mathrm{d}\tau_k$). The importance weight is their ratio (Jonsson 2006, eq. 31),

$$w_t^{(k)} = \frac{\mathrm{e}^{-s_t \tau_k} \, s_t}{\mathrm{e}^{-\tau_k}} = s_t \, \mathrm{e}^{(1-s_t)\,\tau_k}, \tag{9}$$

and the cumulative per-photon optical-depth weight after k segments is

$$W_\tau^{(k)}(t) = \prod_{j=1}^{k} w_t^{(j)} = s_t^k \, \exp\Big[(1 - s_t) \textstyle\sum_{j=1}^{k} \tau_j\Big]. \tag{10}$$

The forced first segment uses the identical factor (9); the $(1 - \mathrm{e}^{-\tau^{\rm e}})$ normalization of forced scattering cancels in the contribution ratio, as shown in Appendix A.

3.2 Peel attenuation and the escape decision

The optical depth from a scattering site to the observer also scales, so the extinction factor in the peel is evaluated per target,

$$\mathrm{e}^{-\tau_{\mathrm{obs}}(P_k)} \longrightarrow \mathrm{e}^{-s_t\,\tau_{\mathrm{obs}}(P_k)}. \tag{11}$$

A useful property of the grey rescaling is that the *interact-versus-escape* decision is scale invariant: a non-forced segment draws $\tau_k = -\ln R$ and escapes if τ_k exceeds the edge optical depth D_k; since the target edge optical depth is $s_t D_k$ and the target interaction depth would be $s_t \tau_k$, the ratio τ_k/D_k is independent of s_t. Hence the entire forward walk is driven by the reference run, and the scan only (i) accumulates W_τ and (ii) applies the per-target attenuation (11). The random-number stream is untouched, so the slice at $\tau_t = \tau_0$ (i.e. $s_t = 1$) is bit-for-bit identical to an ordinary single run.

4. Combined five-dimensional estimator

Stacking the albedo, asymmetry, and optical-depth reweightings, the k-th scattering deposits, for every grid point (a_i, g_l, τ_t),

$$J(x,y,a_i,g_l,\tau_t) \mathrel{+}= \underbrace{\frac{w}{r_k^2}}_{\text{base}} \underbrace{A_{a_i}^{(k)}}_{\text{albedo}} \underbrace{\frac{\Phi(\mu_{\mathrm{obs},k}|g_l)}{4\pi} W_{g_l}^{(k-1)}}_{\text{asymmetry}} \underbrace{\mathrm{e}^{-s_t\tau_{\mathrm{obs}}(P_k)} W_\tau^{(k)}(t)}_{\text{optical depth}}. \tag{12}$$

The three parameter axes never mix inside the bracketed factors, so (12) is the outer product of three vectors scaled by the common geometric base w/r_k^2. Defining

$$u_a(i) = A_{a_i}^{(k)}, \qquad v_g(l) = \frac{\Phi(\mu_{\mathrm{obs},k}|g_l)}{4\pi} W_{g_l}^{(k-1)}, \qquad q_\tau(t) = \mathrm{e}^{-s_t\tau_{\mathrm{obs}}(P_k)} W_\tau^{(k)}(t), \tag{13}$$

the per-event update is the rank-one tensor $J_{ilt} \mathrel{+}= (w/r_k^2)\, u_a(i)\, v_g(l)\, q_\tau(t)$. The cost is therefore one ray trace and $n_a n_g n_t$ multiply–add operations per scattering per observer, while the forward random walk is unchanged.

Direct (unscattered) light. Unlike a and g, the optical depth changes the attenuation of the direct beam. With w_0 the emission weight and $\tau_{\mathrm{obs}}^{\mathrm{src}}$ the source-to-observer optical depth, the direct image is

$$J_{\mathrm{dir}}(x,y,\tau_t) \mathrel{+}= \frac{w_0}{4\pi r^2}\,\mathrm{e}^{-s_t\,\tau_{\mathrm{obs}}^{\mathrm{src}}}, \tag{14}$$

which carries a τ axis but no a or g axis. The unattenuated image $J_{\mathrm{dir},0} = w_0/(4\pi r^2)$ is independent of all three parameters and stays two-dimensional.

5. Implementation in MoCafe

The scan state lives in `scan_mod.f90`. The per-photon accumulators A_a (`scan_Aalb`), W_g (`scan_Wg`), and W_τ (`scan_Wtau`) are reset at each photon birth. Per scattering event: A_a and W_τ are advanced *before* the peel (so the k-th peel sees a^k and the product of k optical-depth factors), and W_g is advanced *after* the peel using the actual scattering cosine (so the peel sees the previous $k-1$ scatterings). The free-path optical depth τ_k of the segment that just ended is handed to the scan through the module variable `scan_tau_step`, set in both photon-loop drivers in `run_simulation_mod.f90` immediately before the call to the scatterer.

The switches and grids are

parameter	meaning	default
`par%use_ag_list`	enable the (a,g) scan	`.false.`
`par%albedo_list`	albedo grid $\{a_i\}$	$0.1,\ldots,1.0$
`par%hgg_list`	asymmetry grid $\{g_l\}$	$0.0,\ldots,0.9$
`par%hgg`	simulated g_0	—
`par%use_tau_list`	enable the τ scan	`.false.`
`par%tau_list`	optical-depth grid $\{\tau_t\}$	$\sqrt{2}$-spaced in $[0.5,2]\,\tau_0$
`par%taumax`	simulated τ_0	—

The two switches compose: the scattered image is $J(x,y)$, $J(x,y,a,g)$, $J(x,y,\tau)$, or $J(x,y,a,g,\tau)$ for the four combinations. When the τ scan is on without the (a,g) scan, the a and g axes collapse to length one (holding `par%albedo` and `par%hgg`). The reference τ_0 slice is auto-inserted into `tau_list` if absent. The scan is restricted to the no-Stokes Henyey–Greenstein path and (for τ) to internal sources.

6. Accuracy and variance

All reweightings are unbiased in expectation, but their variance grows as the target parameters depart from the simulated values (Jonsson 2006, §3.9). For the optical-depth scan, the per-segment weight (9) is bounded for $s_t > 1$ (thicker targets) but the forced first segment's second moment, $\propto \int_0^{\tau^{\rm e}} {\rm e}^{(1-2s_t)\tau_1}\,{\rm d}\tau_1$, diverges with $\tau^{\rm e}$ once $s_t < \frac{1}{2}$. Practical guidance:

- keep `tau_list` within a factor of ~ 2–3 of `par%taumax`, and choose `par%taumax` near the (logarithmic) middle of the target range;
- keep $g_0 \simeq 0.4$–0.5 (Seon 2010), and $|g-g_0| \lesssim 0.5$–0.6;
- always include the reference τ_0 (done automatically) for a zero-variance, bit-identical anchor slice.

The implementation has been verified on a point source in a uniform sphere: the $s_t = 1$ scattered slice is bit-identical to a single run (and the direct beam is exact at every τ_t since it involves no Monte-Carlo); the off-reference scattered slices reproduce independent single-τ runs to $\lesssim 0.15\%$ in integrated flux, with residual pixel noise growing toward the faint wings, as expected.

A Cancellation of the forced-scattering normalization

We show that the forced first segment uses the same weight (9) as an ordinary segment, i.e. that the $(1-{\rm e}^{-\tau^{\rm e}})$ factor cancels in the contribution ratio.

For the reference run the forced first scattering multiplies the packet weight by $w = (1-{\rm e}^{-\tau^{\rm e}})$ and samples τ_1 from the density f_1 of (2). Writing $g(\tau_1)$ for the (parameter-dependent) peel factors that multiply the weight at the interaction point, the reference image expectation per emitted photon is

$$E_{\rm ref} = \int_0^{\tau^{\rm e}} f_1(\tau_1)\, w\, g(\tau_1)\,{\rm d}\tau_1 = \int_0^{\tau^{\rm e}} {\rm e}^{-\tau_1}\, g(\tau_1)\,{\rm d}\tau_1, \tag{15}$$

where the normalization $(1-e^{-\tau^e})$ in w cancels the same factor in the denominator of f_1.

For an independent run at scaling s (target optical depth $s\tau_0$), the edge depth is $s\tau^e$ and the forced first scattering samples τ_1' from $e^{-\tau_1'}/(1-e^{-s\tau^e})$ with weight $(1-e^{-s\tau^e})$. Substituting $\tau_1' = s\tau_1$ (so that the physical interaction point, hence g, coincides with the reference's) gives

$$E_{\rm indep}(s) = \int_0^{\tau^e} e^{-s\tau_1}\, s\, g_s(\tau_1)\, d\tau_1, \tag{16}$$

where g_s differs from g only through the per-target peel attenuation $e^{-s\tau_{\rm obs}}$ already accounted for in (11). Reweighting the reference samples by (9) reproduces this exactly:

$$\int_0^{\tau^e} e^{-\tau_1}\, g_s(\tau_1) \left[s\, e^{(1-s)\tau_1} \right] d\tau_1 = \int_0^{\tau^e} s\, e^{-s\tau_1}\, g_s(\tau_1)\, d\tau_1 = E_{\rm indep}(s). \tag{17}$$

The factor $(1-e^{-\tau^e})$ never appears, so the forced first segment is weighted by $s\,e^{(1-s)\tau_1}$, identical in form to an ordinary (non-forced) segment. The same change of variables, applied segment by segment, establishes the unbiasedness of (10) for all scattering orders. ■

References

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